

THE EXACT CEILING OF THE PRODUCT RELAXATION FOR THE UNION-CLOSED SETS CONJECTURE

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ABSTRACT. Let c_{UC} be the largest universal lower bound on the maximal element frequency of a finite union-closed family. Gilmer’s entropy method, as developed by Liu and Costa, proceeds by combining an independent construction with conditionally i.i.d. ones and relaxing each through a product minorant. We show that this relaxation has an exact ceiling: no choice of weights and no number of conditionally i.i.d. constructions can yield more than

$$\hat{c} = \frac{1}{2} - \max_x \frac{xH(x) - \frac{1}{2}H(x^2)}{\log 2} = 0.3828852599667978\dots$$

We then exhibit a certificate within $6.6 \cdot 10^{-7}$ of this ceiling, giving $c_{\text{UC}} \geq 0.3828846$, obtained by tuning the coupling parameter of Liu’s Example 5 to a tangency condition. The ceiling bounds the product relaxation, not Liu’s criterion in full generality.

1. INTRODUCTION

A finite family $\mathcal{F}$ of sets is *union-closed* if $A \cup B \in \mathcal{F}$ whenever $A, B \in \mathcal{F}$. Frankl’s conjecture asserts that every such family containing a nonempty set has an element belonging to at least half of its members; see the survey [2]. Writing

$$\text{freq}(\mathcal{F}) := \max_{i \in \bigcup \mathcal{F}} \frac{|\{A \in \mathcal{F} : i \in A\}|}{|\mathcal{F}|}, \quad c_{\text{UC}} := \inf_{\mathcal{F}} \text{freq}(\mathcal{F}),$$

the conjecture is equivalent to $c_{\text{UC}} = 1/2$; the family $\{\emptyset, \{1\}\}$ gives $c_{\text{UC}} \leq 1/2$. Throughout $\log$ is the natural logarithm and $H(x) := -x \log x - (1-x) \log(1-x)$, $0 \log 0 = 0$.

Gilmer [6] proved $c_{\text{UC}} \geq 0.01$; the sharp one-coordinate i.i.d. inequality [4, 1, 8] gives $c_{\text{UC}} \geq (3 - \sqrt{5})/2$, improved to about 0.3823455 by [9, 3]. Liu [7] introduced conditionally i.i.d. couplings and conjectured a threshold c_L ; Costa [5] proved $c_{\text{UC}} \geq c_L$ and then $c_{\text{UC}} \geq c_L + 2 \cdot 10^{-5} > 0.382729087918735$.

All of these arguments instantiate Liu’s criterion with an independent protocol $\Pi^{(0)}$ of weight w_0 together with conditionally i.i.d. protocols $\Pi^{(j)}$ of weights w_j , each of the latter handled through a *product minorant*

$$(1) \quad H(\Pi_{x,y}^{(j)}(0,0)) \geq \psi_j(x)\psi_j(y).$$

Our first result is that this device has an exact and computable ceiling.

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Theorem 1.1. *Let $w_0, w_j \geq 0$ with $w_0 + \sum_j w_j = 1$ and suppose each conditionally i.i.d. protocol is relaxed as in (1). If the resulting criterion certifies the value $c = 1 - m$, then*

$$m \geq \hat{m} := \frac{1}{2} + \max_{0 \leq x \leq 1} \frac{xH(x) - \frac{1}{2}H(x^2)}{\log 2} = 0.617114740033202\dots,$$

that is, $c \leq \hat{c} = 1 - \hat{m} = 0.3828852599667978\dots$

Proof. Probe with the point mass $P = \delta_x$, admissible for $x \geq m$. Feasibility requires

$$(2) \quad m \left[w_0 H(x^2) + \sum_j w_j \psi_j(x)^2 \right] \geq xH(x).$$

Putting $y = x$ in (1) gives $\psi_j(x)^2 \leq H(\Pi_{x,x}^{(j)}(0,0)) \leq \log 2$, since $H \leq \log 2$ pointwise; note this uses nothing about the particular construction.

Next probe with $P = (1 - \varepsilon)\delta_1 + \varepsilon\delta_y$ and let $\varepsilon \rightarrow 0$. At $x = y = 1$ every protocol has $\Pi(0,0) = 1$, hence $H = 0$ and $\psi_j(1) = 0$ by (1); so each relaxed term contributes at order ε^2 , while the independent term retains $H(xy) = H(y)$ at order ε . Comparing the ε -coefficients forces

$$(3) \quad w_0 \geq \frac{1}{2m}.$$

Since $\hat{x} := \arg \max_x [xH(x) - \frac{1}{2}H(x^2)]$ satisfies $H(\hat{x}^2) = 0.6921211 < \log 2$, the bound (2) is weakest at $w_0 = 1/(2m)$, where it reads $\frac{1}{2}H(x^2) + (m - \frac{1}{2})\log 2 \geq xH(x)$ for every x . Rearranging and maximising over x gives $m \geq \hat{m}$. $\square$

Remark. Constraint (3) is already present in [5], in a form its author does not exploit. Since $g(1) = 1$, the term $\sqrt{H(g(x))H(g(y))}$ in [5, Lemma 3.2] vanishes identically on the edge $y = 1$, where that lemma reduces to $2\alpha H(x) \geq H(x)/m_*$, i.e. to $\alpha \geq 1/(2m_*)$ — exactly (3). At Costa's parameters $\alpha = 0.899947 > 0.809991 = 1/(2m_*)$, a slack of 0.0900. Theorem 1.1 identifies that slack as the entire remaining budget of the method: closing it moves c from 0.382729 to $\hat{c}$, a gain of $1.562 \cdot 10^{-4}$ and no more.

We stress that (3) is derived above from the *integrated* criterion, by the ε -probe, and not from a pointwise inequality. This matters: a pointwise bound such as [5, Lemma 3.2] is a sufficient condition, strictly stronger than what the criterion demands, so a constraint read off its boundary need not bind in general. The probe shows (3) binds for *every* instantiation of the shape considered here, however the integration is performed.

Remark. Theorem 1.1 bounds the product relaxation (1), not Liu's criterion itself: the exact infimum $\inf_{\rho \in K_j(\mu)}$, or protocols outside the class C_3 , are not covered. It explains, however, why adding further constructions cannot help. Liu's Example 4 is chosen to maximise $H(\Pi_{s,t}(0,0))$ for given t , so it already saturates $\log 2$ on the diagonal; and adding a third construction of weight θ

moves (3) to $(1-\theta)w_0 \geq 1/(2m)$ while gaining only $m\theta(\varphi(\hat{x})^2 - \log 2)$ there, which for Costa's φ is a loss, $\varphi(\hat{x})^2 \approx \log 2(1 - 2 \cdot 10^{-5})$.

Our second result is a certificate essentially at the ceiling. For $l \in [0, 1]$ set

$$r_l(x, y) := xy + l^2x(1-x)y(1-y), \quad g_l(x) := r_l(x, x) = x^2(1 + l^2(1-x)^2),$$

so $l = 1$ is the construction of [5].

Theorem 1.2. $c_{UC} \geq 0.3828846$, within $6.6 \cdot 10^{-7}$ of $\hat{c}$.

The binding constraints are (3) and a single interior contact, both governed by $H(g_l(\hat{x}))$, which is largest when its argument is $1/2$. This fixes l by the *tangency condition*

$$(4) \quad g_l(\hat{x}) = \frac{1}{2}, \quad l^2 = \frac{1/(2\hat{x}^2) - 1}{(1 - \hat{x})^2}, \quad l = 0.70468061383263156 \dots$$

We work at the rational $l = 1409/2000$, $t := l^2 = 1985281/4000000$, so that Section 3 is exact, and verify the criterion at

$$m = 0.6171154, \quad \eta = \beta(m) - \delta_\eta, \quad \beta(m) := 1 - \frac{1}{2m}, \quad \delta_\eta = 3 \cdot 10^{-7},$$

with weights $\alpha = 1 - \eta$, $\beta = \eta$ in the sense of [5, §3]. By (3) the choice of η is forced and is *not* the derivative condition [5, (10)]: at $l = 1$ the latter gives $\alpha = 0.899947 > 0.809991 = 1/(2m_*)$, comfortably feasible, whereas at the tuned l it gives $\alpha = 0.809903 < 0.810222 = 1/(2\hat{m})$ and violates (3). Tuning l trades the diagonal derivative condition for saturation of the endpoint constraint; as $\delta_\eta \rightarrow 0$ one recovers exactly $\hat{m}$, which is why Theorem 1.1 is sharp for this family.

2. TWO ENTROPY INEQUALITIES

The two purely entropic ingredients of [5, §4] are independent of l ; we prove them, as they are used here at $l \neq 1$. Substitute $u = 1/(1 + e^t)$, so $t = H'(u)$, and put $\sigma := 1 - u$, $w := e^{-t}$, so that

$$(5) \quad A(t) := H(u) = \log(1+w) + \frac{tw}{1+w}, \quad A'(t) = -t\sigma(1-\sigma), \quad A > 0.$$

Lemma 2.1. $M(u) := uH'(u)/H(u)$ is strictly decreasing on $(0, 1)$.

Proof. As u decreases in t it suffices that $\varphi(t) := t(1-\sigma)/A(t)$ increase. By (5), $\varphi' = (1-\sigma)[(1-t\sigma)A + t^2\sigma(1-\sigma)]/A^2$, so we need $\Psi := (1-t\sigma)A + t^2\sigma(1-\sigma) > 0$. For $t \leq 0$ this holds as $1-t\sigma \geq 1$. For $t > 0$, substituting $\sigma = 1/(1+w)$ and multiplying by $(1+w)^2$,

$$\begin{aligned} (1+w)^2\Psi &= (1+w-t)[(1+w)\log(1+w) + tw] + t^2w \\ &= (1+w)[(1+w)\log(1+w) + t(w - \log(1+w))], \end{aligned}$$

and both bracketed terms are positive since $\log(1+w) < w$. $\square$

Lemma 2.2. For $u \in (0, \frac{1}{2})$, $-\Lambda''(u)/\Lambda'(u) > 1/u + 1/(1-u)$, where $\Lambda = \log H$.

Proof. As $H'' = -1/(u(1-u))$ we have $-\Lambda''/\Lambda' = H'/H + 1/(u(1-u)H')$, so with $H' > 0$ the claim is $u(1-u)H'^2 > H(H'-1)$, i.e. $t^2\sigma(1-\sigma) > A(t)(t-1)$. For $0 < t \leq 1$ the right side is nonpositive. For $t > 1$, multiplying by $(1+w)^2$ and using $\log(1+w) \leq w$ reduces the claim to $t^2 > (t-1)(1+w)(1+w+t) = t^2 - 1 + (t-1)w(2+w+t)$; so it suffices that $(t-1)e^{-t}(2+e^{-t}+t) < 1$, whose maximum over $t > 1$ is 0.57383... $\square$

3. THE POINTWISE INEQUALITIES AT PARAMETER l

Write $W = 1+t(1-x)(1-y)$, $U = 1+t(1-2x)(1-y)$, $V = 1+t(1-x)(1-2y)$; then $r_l = xyW$, $\partial_x r_l = yU$, $\partial_y r_l = xV$, $\partial_{xy} r_l = 1+t(1-2x)(1-2y)$.

Lemma 3.1. $W \partial_{xy} r_l = UV + txy$, hence $\partial_{xy} r_l / (\partial_x r_l \partial_y r_l) = 1/r_l + t/(WUV)$.

Proof. A polynomial identity in x, y, t ; the difference expands to 0. $\square$

At $t = 1$ this is [5, Lemma 4.2]. Following the case division of [5, §4]:

Case A ($g_l(y) < \frac{1}{2}$). There $r_l < \frac{1}{2}$, and the argument of [5, §4] requires $\partial_{xy} r_l / (\partial_x r_l \partial_y r_l) \leq 1/r_l + 1/(1-r_l)$; by Lemma 3.1 this is exactly $WUV \geq t(1-r_l)$, the coefficient t replacing Costa's 1. Since $g_l(y_0) = \frac{1}{2}$ at $y_0 = 0.690917 < 45/64$, it suffices on $[0, 45/64]^2$. Rescaling $x \mapsto \frac{45}{64}x$, $y \mapsto \frac{45}{64}y$, the polynomial $WUV - t(1-r_l)$ of bidegree $(3, 3)$ has all 16 Bernstein coefficients positive, with minimum the exact rational

$$\frac{4652778828628843621308587}{838860800000000000000000} = 0.554654459\dots$$

Case B ($g_l(y) \geq \frac{1}{2}$). Here $\partial_y r_l / r_l = A_x(y)M(r_l)$ with $A_x(y) = V/(yW) = 1/y - t(1-x)/(1+t(1-x)(1-y))$, which is [5, (28)] at $t = 1$. As $u \mapsto tu/(1+tu(1-y))$ is strictly increasing for $u \geq 0$ and every $t > 0$, and $1-x > 1-y$, we get $0 < A_x(y) < A_y(y)$; the rest uses only Lemma 2.1. All of this needs $U, V > 0$, which holds since $W \in [1, 1+t]$ and $U, V \in [1-t, 1+t]$ with $1-t = 0.50368 > 0$; at $t = 1$ they may vanish.

4. THE SECOND POINTWISE INEQUALITY

In coordinates $s = \log(1/x)$ put $\mathcal{U}(s) := e^s H(e^{-s})$ and $\mathcal{W}(s) := H(q)/x^2$, $q = x^2(1+t(1-x)^2)$, $x = e^{-s}$. The inequality to be proved is

$$(6) \quad F(s, \sigma) := (1-\eta)\mathcal{U}(s+\sigma) - \frac{\mathcal{U}(s) + \mathcal{U}(\sigma)}{2m} + \eta\sqrt{\mathcal{W}(s)\mathcal{W}(\sigma)} > 0 \quad (s, \sigma > 0).$$

Lemma 4.1. $\mathcal{U}$ and $\mathcal{W}$ are strictly increasing on $(0, \infty)$.

Proof. $\mathcal{U} = H(x)/x$ and $d\mathcal{U}/ds = [H(x) - xH'(x)]/x = -\log(1-x)/x > 0$, using the identity $H(x) - xH'(x) = -\log(1-x)$. For $\mathcal{W} = H(q)/x^2$ one

computes $d\mathcal{W}/ds = [2H(q) - xq'H'(q)]/x^2$ and, using $xq' = 2q - 2tx^3(1-x)$ and the same identity at q ,

$$2H(q) - xq'H'(q) = -2\log(1-q) + 2tx^3(1-x)H'(q).$$

If $q \leq \frac{1}{2}$ both terms are nonnegative and the first is positive. If $q > \frac{1}{2}$ then $H'(q) < 0$ and $\log(q/(1-q)) \leq -\log(1-q)$ because $\log q \leq 0$; since $t \max_x x^3(1-x) = 27t/256 = 0.0523 < 1$, the first term dominates. $\square$

Write $u = s + \sigma$, assume $s \leq \sigma$, and set $\delta_\eta = (1-\eta) - 1/(2m) > 0$ and $D := \mathcal{U}(s) + \mathcal{U}(\sigma) - \mathcal{U}(u)$. Then (6) reads exactly

$$(7) \quad F = \delta_\eta \mathcal{U}(u) - \frac{D}{2m} + \eta \sqrt{\mathcal{W}(s)\mathcal{W}(\sigma)}.$$

Put $\phi(s) := \mathcal{U}(s) - s$. Writing $z = e^{-s}$ one has $\phi = (1-z)(-\log(1-z))/z = 1 - z/2 - z^2/6 - \dots$, so ϕ is strictly increasing with $0 < \phi < 1$; consequently $\mathcal{U}(u) \geq \max(\mathcal{U}(s), \mathcal{U}(\sigma))$ gives

$$(8) \quad 0 \leq D = \phi(s) + \phi(\sigma) - \phi(u) \leq \min(\phi(s), \phi(\sigma)) \leq 1.$$

Let

$$\rho := \inf_{y>0} \frac{\mathcal{W}(y)}{\mathcal{U}(y)} = 1.612119785, \quad K := \left(\frac{1}{2m\eta\rho}\right)^2 = 7.013240853,$$

the infimum being attained at $y = 0.25$ (the ratio tends to 2 as $y \rightarrow 0$ and to $2(1+t)$ as $y \rightarrow \infty$).

Proposition 4.2. $F > 0$ on $(0, \infty)^2$.

Proof. Four exhaustive cases.

(I) $\mathcal{U}(\sigma)/\mathcal{U}(s) \geq K$. By Lemma 4.1, $\mathcal{U}(u) \geq \mathcal{U}(\sigma)$, so $D \leq \mathcal{U}(s)$ by (8). With $\mathcal{W} \geq \rho\mathcal{U}$,

$$F \geq \delta_\eta \mathcal{U}(u) + \mathcal{U}(s) \left[\eta\rho\sqrt{\mathcal{U}(\sigma)/\mathcal{U}(s)} - \frac{1}{2m} \right] \geq \delta_\eta \mathcal{U}(u) > 0,$$

the bracket being nonnegative by the definition of K . This case is valid at all scales.

In the remaining cases $\mathcal{U}(\sigma) < K\mathcal{U}(s)$, which forces $\sigma \leq 15.575s$.

(II) $s, \sigma \geq 2$. By (8), $D \leq 1$, while $\mathcal{W}$ increasing gives $\mathcal{W}(s)\mathcal{W}(\sigma) \geq \mathcal{W}(2)^2 > (1/(2m\eta))^2 = 18.227$, so $\eta\sqrt{\mathcal{W}\mathcal{W}} > 1/(2m) \geq D/(2m)$ and $F > 0$.

(III) $s, \sigma \leq S := 2^{-10}$. Then $u \leq 2S$ and, with $z = s/u \geq 0.137194$ forced by the balance condition, $\sqrt{z(1-z)} \geq 0.344053$ and

$$2\eta\sqrt{z(1-z)}\log(1/(2u)) \geq 0.1305877 \cdot \log(1/(4S)) = 1.2894 \cdot \frac{\log 2}{2m} \geq \frac{H(z)}{2m},$$

a margin of 28.9% over the required bound.

(IV) *The remainder.* Not (II) gives $s < 2$; not (III) gives $\sigma > S$; with $\sigma \leq 15.575s$ this yields $s > S/15.575 = 2^{-13.97}$ and $\sigma \leq 15.575 \cdot 2 < 2^5$. Hence $(s, \sigma) \in [2^{-14}, 2^5]^2$, where (6) is verified by subdivision as described in Section 5. $\square$

Proposition 4.2 together with Section 3 verifies the criterion at (l, m, η) , proving Theorem 1.2.

5. THE COMPUTATION IN CASE (IV)

Because $\mathcal{U}$ and $\mathcal{W}$ are increasing (Lemma 4.1), on a box $[s_1, s_2] \times [\sigma_1, \sigma_2]$ the needed one-sided bounds are attained at corners: one needs $\mathcal{U}$ from below at $s_1 + \sigma_1$, from above at s_2 and σ_2 , and $\mathcal{W}$ from below at s_1 and σ_1 . Setting

$$D_{\text{lo}} := (1 - \eta)\mathcal{U}^-(s_1 + \sigma_1) - \frac{\mathcal{U}^+(s_2) + \mathcal{U}^+(\sigma_2)}{2m},$$

a box is accepted when $D_{\text{lo}} \geq \tau$ or $\eta^2\mathcal{W}^-(s_1)\mathcal{W}^-(\sigma_1) - \max(0, -D_{\text{lo}})^2 \geq \tau$, with $\tau = 10^{-10}$; boxes that fail are bisected in both coordinates.

Outward rounding. $\mathcal{U}^\pm$ and $\mathcal{W}^-$ are not bare floating-point values. Each expression is evaluated in IEEE double precision in the three numerically stable regimes

$$\mathcal{U}(s) = s + \varepsilon(-\log \varepsilon)e^s \quad (s < 1), \quad \mathcal{U}(s) = s + (e^s - 1)(-\log(1 - e^{-s})) \quad (1 \leq s < 700), \quad \mathcal{U}(s) = s + 1$$

$\varepsilon = -\text{expm1}(-s)$, and analogously for $\mathcal{W}$; the result is then widened outward by the relative pad $2^{-45} = 2.842 \cdot 10^{-14}$, about 128 units in the last place. This yields rigorous enclosures provided the accumulated relative error of the evaluations stays below 2^{-45} on the domain.

Validation of the pad. That hypothesis was checked against `mpmath`'s correctly outward-rounded interval arithmetic at 40 digits, on 8037 points of $[2^{-14}, 2^5]$ comprising a geometric mesh, a random sample, and all dyadic corners and midpoints used by the subdivision. The padded enclosures contained the rigorous intervals in every case (no failures), and the observed relative error of the double-precision evaluation was at most $4.41 \cdot 10^{-16}$, roughly two units in the last place; the pad therefore carries a factor 64 of headroom, and the acceptance threshold $\tau = 10^{-10}$ a further factor of about 10^3 over the pad's absolute effect on the domain.

Result. Dyadic subdivision of $[2^{-14}, 2^5]^2$ accepts 168 658 219 boxes, with no box left undecided.

Remark. The computation is therefore outward-rounded rather than bare floating point, but it is not carried out in MPFR throughout, as are the corresponding computations of [5, Appendix A]; it rests on the validated hypothesis above rather than on correct rounding at every step. An MPFR implementation would remove that hypothesis. Cases (I)–(III) are analytic and use no computation beyond the constants ρ , K and the two thresholds.

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